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Invited Paper

Nissan LEAF and e-POWER: Evolution of Motors and Inverters

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Nissan developed LEAF as a mass-produced electric vehicle (EV) in 2010 and has been continuously improving it ever since. In 2016, Nissan also developed NOTE with e-POWER, a 100% motor-driven system, thus delivering excellent 100% motor-driven experience to customers. This paper describes the evolution of the motor and inverter the key components that support Nissan's 100% motor-driven system.

Keywords: electric vehicle, motor, inverter, powertrain integration, NVH, power module thermal control

1. Introduction

To contribute to the prevention of global warming, Nissan launched the mass-produced battery electric vehicle "Nissan LEAF" in 2010 as an electric vehicle, which is practical for many customers, that is, an affordable zero-emission vehicle that can accommodate five people and meet the requirements of daily driving range. Since then, Nissan has continued to promote vehicle electrification. Furthermore, on January 2021, Nissan announced its goal and strategy to become carbon neutral by 2050. (Figure 1) As a part of this effort, by the early 2030s, every new commercially available Nissan vehicle will be electrified. EV and e-POWER will be the two main powertrains[(1)].

This chapter begins with a description of Nissan's proprietary e-POWER system, a unique electrification technology based on battery EV technology. Simultaneously, Nissan, which pioneered the development of battery EVs, recognized the superiority of 100% motorized driving at an early stage. The e-POWER system is the result of Nissan's research and development efforts to bring this wonderful driving experience to as many customers as possible.

Figure 2 shows the drive system configuration of EV and e-POWER. EV and e-POWER share components such as motors, inverters, and gear sets as well as the control logic of the drive system. Specifically, to drive the motor, EV uses electricity stored in a large-capacity battery, while e-POWER uses a small-capacity battery and a dedicated high-efficiency engine as a generator. By sharing the same powertrain components, EV and e-POWER can quickly respond to different markets needs around the world according to their energy and infrastructure readiness, and strongly promote the electrification of automobiles.

Additionally, e-POWER is classified as a hybrid EV (HEV). The main difference to conventional HEV is that,

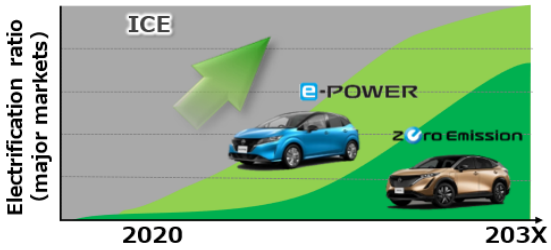


Fig. 1. Nissan electrified load map

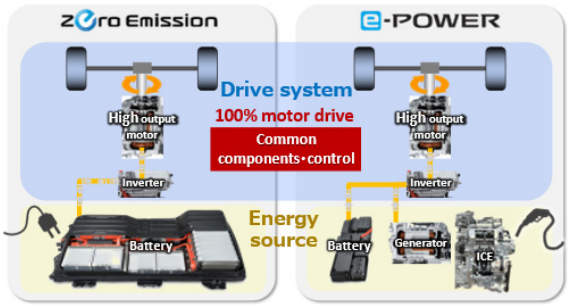


Fig. 2. System configuration of EV and e-POWER

similar to EVs, e-POWER, is 100% motor-driven. A 100% motor-driven car offers a new driving experience that cannot be experienced or achieved with conventional internal combustion engines or HEV based on internal combustion engines. As an example, Figure 3 shows the characteristics of acceleration from a standstill, and the e-POWER provides faster acceleration response and smoother, vibration-free acceleration compared to other HEVs. Since the first e-POWER was launched in the Japanese market in 2016, many customers have supported the 100% motor-driven driving experience, and by the second quarter of 2022, Nissan's 100% motor-driven electrified vehicles will account for 52% of the total sales[(2)]. (Figure 4)

Today, many automotive manufactures entered the market of the battery EV, and the various electric powertrains were introduced with the evolution of technologies for electric motors and power-electronics. Regarding the development of electric components used for HEV, Okamura summarized the

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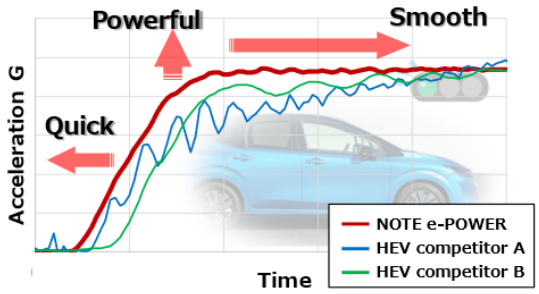


Fig. 3. Acceleration at standing start

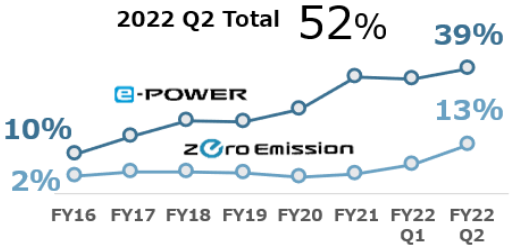


Fig. 4. 100% motor drive vehicle sales ratio in Japan

evolution of electric components in Prius[3]. As the review papers, the specifications or the general knowledge of battery EVs and HEVs were shown in (4)–(6). However, it is difficult to find the technical literatures which describe the evolution of the electric components used for battery EVs or other 100% electric drive systems.

This paper presents the evolution of motors and inverters for EVs and e-POWERs that support Nissan's electrification strategy, and describe not only electric technology evolution but also mechanical and structural design innovations on motors and inverters.

2. The Evolution of Nissan EV and e-POWER

2.1 Electrification Vehicle Evolution Table 1 shows Nissan's EV and e-POWER lineup, the appearance of the electric-powertrain (ePT), and the motor/inverter generations. In 2010, Nissan launched the mass-produced EV 2011 model year (11MY) LEAF, and in 2012, the ePT was integrated into a form similar to that of a conventional engine by integrating the motor, inverter, power delivery module (PDM), and reduction gear into a single unit, thereby improving ease of vehicle installation and development efficiency; Furthermore, the model lineup was expanded to include the e-NV200 minivan-type vehicle.

In 2018, Nissan expanded its electric vehicle model to the SYLPHY sedan-type vehicle by evolving the motor and inverter components, while retaining the basic structure. For silent operation and low power consumption at high speeds, the ARIYA mid-size sports utility vehicle (SUV) uses an external excited synchronous motor (EESM), which is different from the interior permanent magnet synchronous motor (IPMSM) that used LEAF, however, the inverter module, a key component of the inverter, has the same structure as the e-POWER.

The e-POWER, on the other hand, was introduced to the Japanese market in 2016 as the NOTE e-POWER as an additional version of the existing internal combustion engine (ICE) vehicle, the NOTE. This e-POWER shares the same motor and inverter as the 18MY LEAF and offers the same 100% motor-driven ride as Nissan's EV. The e-POWER model was later expanded to include the SERENA minivan and the KICKS compact SUV.

Additionally, Nissan has further downsized the unit by integrating the inverter, which was previously mounted on the body, into the motor in 2020. The ePT is now available in e-POWER-only NOTE/AURA compact hatchbacks, KICKS compact SUV and SYLPHY sedan[7]–[14]

As mentioned in the previous section, Nissan has shared the main components of motors, inverters and reduction gears for EVs and e-POWERs, while updating them in line with technological evolution and promoting component integration to reduce the size and weight of ePT.

Table 1. The evolution of Nissan EV/e-POWER

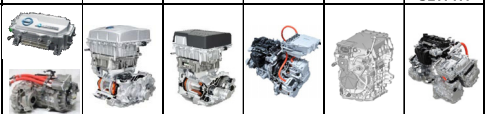
	EV		EV	e-POWER	EV	e-POWER
Vehicle	11MY LEAF	13MY LEAF, e-NV200	18MY LEAF, SLYPHY EV	16MY NOTE, SERENA, KICKS	ARIYA	20MY NOTE, AURA, KIKCS, SLYPHY
e-Powertrain						
Inverter	1st generation		2nd generation		3rd generation	
Motor	1st gene. IPMSM	2nd generation IPMSM			EESM	3rd gene. IPMSM

Table 2. Specification of Nissan EV

Vehicle	11MY LEAF	13MY LEAF, e-NV200	18MY LEAF, SLYPHY EV	18MY LEAF e+	ARIYA
Inverter	1st generation		2nd generation		3rd gene.
Motor	1st gene. IPMSM	2nd generation IPMSM			EESM
Battery	EV 24kWh		EV 30/40kWh	EV 60kWh	EV 60kWh /91kWh
Max. Torque	280Nm	254Nm	320Nm	320Nm	300Nm
Max. Power	80kW	80kW	110kW	160kW	160kW /178kW

2.2 Specification of EV and e-POWER Table 2 and Table 3 show the basic specifications of EVs and the basic specifications of e-POWER.

The second-generation motor was introduced in the 13MY LEAF. With the evolution of the inverter from the first generation to the second, the motor control system evolved and the output power increased due to the increase in battery capacity, the output power of the second-generation motor almost doubled, from 80 kW in the 13MY LEAF to 160 kW in the 18MY LEAF e+.

Table 3. Specification for Nissan e-POWER

Vehicle	16MY NOTE / KICKS	SERENA	20MY NOTE	AURA / KIKCS / SLYPHY
Inverter	2nd generation		3rd generation	
Motor	2nd generation IPMSM		3rd generation IPMSM	
Battery	HEV 80Cell	HEV 96Cell	HEV 80Cell	HEV 96Cell
Max. Torque	254Nm	320Nm	280Nm	300Nm
Max. Power	80kW	100kW	85kW	100kW

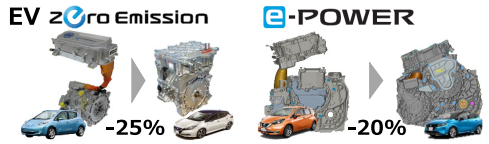


Fig. 5. Volume reduction ratio by ePT integration

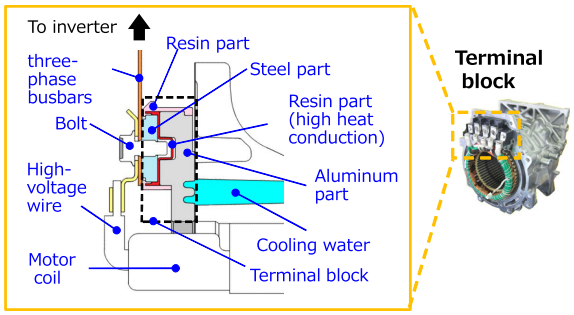


Fig. 6. Cross-section of the terminal block in the motor

On the other hand, e-POWER uses the same engine dedicated to power generation within the generation. However, the performance of the system as a whole has been improved and the range of applications has been extended to larger vehicles, such as the Serena minivan.

3. Integration of e-powertrain

Major electrical components have been integrated to improve package efficiency, increase cabin size and simplify vehicle installation for both EV and e-POWER. The volume of the ePT was reduced by approximately 25% from the first-generation 11MY LEAF to the second-generation 13MY LEAF, and by approximately 20% from the first-generation e-POWER 16MY NOTE to the second-generation e-POWER 20MY NOTE. (Figure 5) The integration of ePT components is a very important technology for reducing the size and weight of ePTs. However, there are following four major challenges to achieve this integration.

1. Thermal isolation structure between motor and inverter
2. Layout of internal inverter components to downsize the ePT
3. Recovery from noise and vibration performance degradation
4. Improvement of vibration resistance of electronic components

The efforts made by Nissan regarding the four points mentioned above will be described by means of specific examples.

3.1 Thermal Isolation between Motor and Inverter

Under normal operating conditions, the motor and inverter are used in a state where the motor side gets hot. In the first-generation LEAF, the motor and inverter were connected by a high-voltage harness, so the heat generated on the motor side was dissipated by the high-voltage harness along the way and had little effect on the inverter. In the 13MY LEAF, to simplify vehicle installation, the motor, inverter, and PDM have been integrated into a single unit. The motor and inverter are directly connected by a 3-phase bus bar without a high-voltage wiring harness. Therefore, the heat generated on the motor side is transferred to the inverter through the 3-phase busbar, affecting the inverter components, especially the current sensor on the inverter busbar, causing the current sensor temperature to rise to its limit.

As shown in Figure 6, the base of the motor terminal block connecting the motor and inverter is made of aluminum with high thermal conductivity, and cooling water is directly applied to improve cooling performance. Additionally, the resin used to insulate the high-voltage section from the case ground is made of a high thermal conductivity material and is made as thin as possible to reduce thermal resistance. This terminal block structure effectively cools the heat generated by the motor windings and reduces the rate of rise in temperature of the current sensor.

3.2 Layout of Internal Inverter Components to Downsize the ePT

High-voltage harnesses connecting components occupy a large volume in the engine compartment because the harness wires are thick and can only be bent with large curvatures, and the amount of swing corresponding to the relative motion between components must be secured. Therefore, integration of ePT can significantly reduce the volume of ePT in the engine compartment, even if the size of each component is not reduced. As mentioned in the previous section, to reduce the thermal influence of the motor on the current sensor located on the 3-phase busbar, it is effective to increase the distance between the motor and the inverter power module and increase the heat dissipation area from the 3-phase busbar. For this reason, in the 13MY LEAF, the power module in the inverter is placed on the upper side of the vehicle opposite to the motor to increase the length of the 3-phase busbar and improve the heat dissipation performance, whereas, simultaneously, capacitor and other tall components are placed in the center of the motor length to reduce the overall height of the unit. (Figure 7)

3.3 Noise and Vibration Performance Improvement

Generally, the ePT integration has more components than the motor alone, therefore the vibration of the motor is transmitted to more parts that produce noise, then the noise and vibration performance deteriorates. Figure 8 shows the noise measurement results of 11MY LEAF and 13MY LEAF. The initial design of 13MY LEAF shows that the noise is significantly worse compared to that of 11MY LEAF at 1.7 kHz

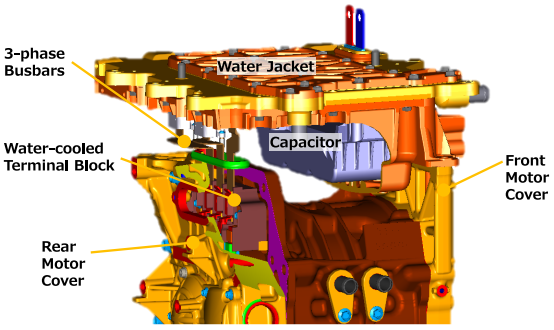


Fig. 7. Positional relation of terminal block, current sensors and water jacket

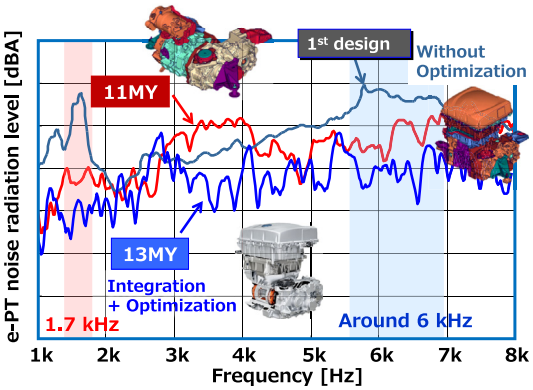


Fig. 8. Noise measurement result of ePT

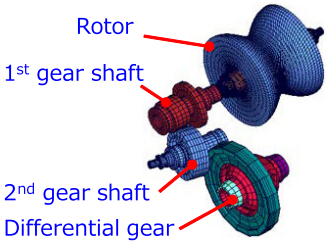


Fig. 9. Vibration mode of rotor and gear system

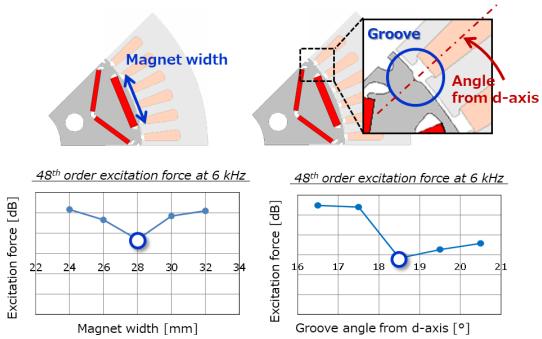


Fig. 10. Electromagnetic force reduction

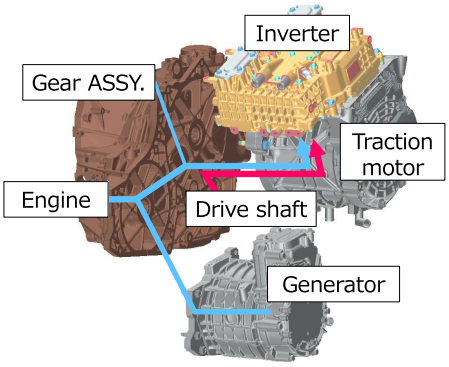


Fig. 11. ENG vibration transfer path to inverter

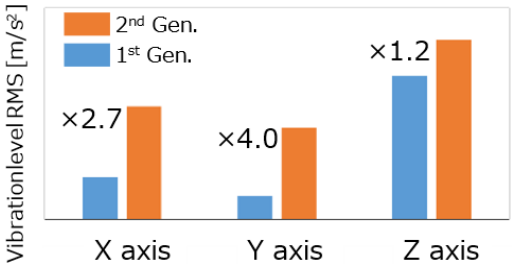


Fig. 12. Vibration level at inverter

and 6 kHz. Investigation showed that the noise at 1.7 kHz is caused by the double resonance of the rotor torsion resonance frequency, which is caused by the two-part skew of the rotor, and gear resonance. (Figure 9) Raising the torsional resonant frequency of the rotor to 2.2 kHz and above resulted in a −20 dB reduction in the 1.7 kHz noise. The 6 kHz noise is found to be generated by the 48-order electromagnetic excitation force of the motor rotation. Therefore, the design of the magnetic circuit, which is the source of the electromagnetic excitation force, was reviewed. Specifically, as shown in the Figure 10, the width of the magnets and the position of the grooves on the rotor surface were optimized to reduce the 48th-order rotational electromagnetic excitation force. As a result, the 13MY LEAF, achieves the same or better noise and vibration performance as the 11MY LEAF with separate ePT structure, despite its integrated ePT structure.

3.4 Improvement of Vibration Resistance of Electronic Components The integration of ePT into e-POWER is even more challenging because it has to withstand the large vibrations caused by the combustion pressure of the engine. As shown in Figure 11, the excitation force caused by the combustion engine is transmitted to the inverter through the gearbox and the drive motor. Figure 12 shows the vibration level of the e-POWER inverter in each direction. Particularly, the X and Y directions are approximately 3 to 4 times greater than those of the 1st generation e-POWER. This is because the inverter is offset from the motor, which causes local resonance phenomena in the inverter part. In terms of vibration resistance characteristics, among the electronic parts of inverter components, the leads of ICs, electrolytic capacitors, connectors, and other parts are particularly susceptible to vibration input, and the vibration resistance design of the leads has a significant impact on the inverter quality.

Figure 13 shows the structure of the third-generation inverter module used in the second-generation e-POWER. The third-generation inverter has a modular structure that integrates the power module, drive board and motor control board. In the vibration test, when the excitation force described above is applied to this inverter module, resonance of the signal pins occurs around 650 and 850 Hz. This is because of the resonance between the power module at the bottom and the electronic board at the top. Signal pin B, which consists of multiple signal pins, has high rigidity and

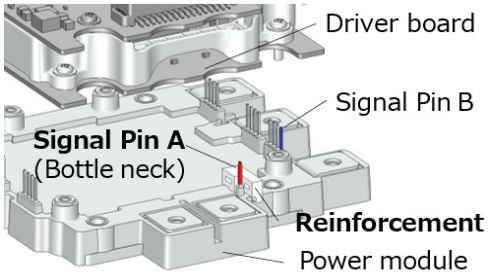


Fig. 13. Weak parts of power module for vibration

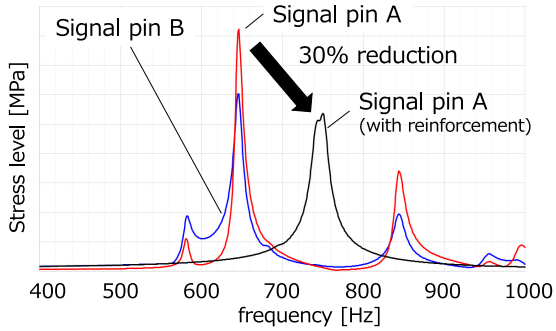


Fig. 14. Freq. characteristic of signal pin on Power module

low stress, while signal pin A, which consists of a single pin, has low rigidity and high stress. The resonance frequency was increased to 750 Hz and the stress generated was reduced by approximately 30% by adding a component to increase the rigidity of signal pin A. (Figure 14)

Consequently, by integrating ePT, Nissan has solved various challenges for both EV and e-POWER, and achieved a compact size.

4. Evolution of Motor

Table 4 shows the motor specifications. From the first-generation to the third-generation, the basic motor specifications, i.e., motor outer diameter, number of poles, and number of stator slots, have been fixed at φ200 mm, eight poles, and 48 slots. The downsizing and weight reduction of the motors was achieved by reducing the motor lamination thickness.

Additionally, the second-generation motor has evolved from the 254 Nm and 80 kW maximum output of the 13MY LEAF, the first vehicle developed, to 340 Nm and 160 kW for the 18MY LEAF e+, owing to advances in the inverter and battery used in combination with the motor.

Until the second-generation, a motor design with a wide range was required because it was the incredibly early stage of electric vehicles. Later, as the electric vehicle segment expanded and the production of 100% motor drive vehicles increased, electric vehicles entered a period of full-scale proliferation. Furthermore, in the third-generation, motors were designed to be optimized for each segment. As a result, the thickness of the 20MY NOTE motor lamination thickness was reduced from 140 to 115 mm, making it the ideal motor for compact cars, and a significant reduction in size and weight was achieved.

The following section describes the technological evolution of the motor.

Table 4. Motor specification

	EV				e-POWER			
	11MY LEAF	13MY LEAF	18MY LEAF	LEAF e+	16MY NOTE	SERENA	20MY NOTE	AURA
	1st gen. IPMSM	2nd generation IPMSM					3rd generation IPMSM	
Max Torque [Nm]	280	254	320	340	254	320	280	300
Max Power [kW]	80	80	110	160	80	100	85	100
Max Motor Speed [rpm]	10,390	10,500		11,330	10,500	11,330	11,330	11,330
Gear Ratio [-]	7.938	8.194		8.194	8.194	8.194	8.194	8.194
MTR Mass [kg]	56.4	54.1			52.3		45.0	
Core diameter [mm]]	φ200	φ200					φ200	
Core length [mm]	150	140					115	
Nb. of turn	8	9					11	
Nb. of coil	15	12					10	
Batt	24kWh	24kWh	30/ 40kWh	62kWh	HEV 80Cell	HEV 96Cell	HEV 80Cell	HEV 96Cell
Max Current [Arms]	400	353	460	500	356	466	404	453

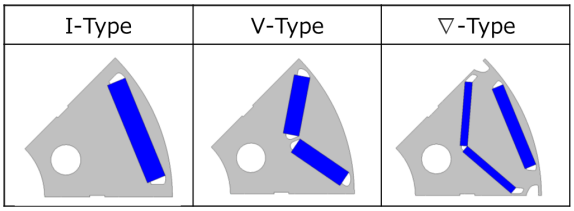


Fig. 15. Rotor shape

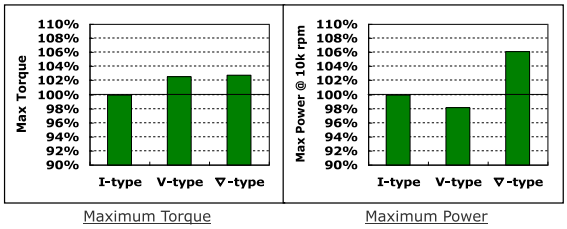


Fig. 16. Output comparison of Rotor shape

Because the speed-torque characteristics of an IPMSM are significantly affected by the magnet arrangement embedded in the rotor, Nissan first studied the speed-torque characteristics of IPMSM with various magnet arrangements at the start of its first EV development.

Figure 16 shows a comparison of the magnet arrangement in the I, V and ▽ rotors, the maximum torque at low speed and the maximum power at maximum speed, this comparison uses the basic geometry (Figure 15), I-type, as a 100% reference, the amount of magnets was kept constant for all the rotors with each type of magnet arrangement, and the stator, winding specifications, and motor lamination thickness were all the same.

In V-type, the rotor core outside the magnet can have a q-axis magnetic path, which allows a larger reluctance torque, resulting in a larger maximum torque compared to that of type I. However, the electromagnetic steel on the outside of the magnet is also a d-axis magnetic path, which causes more magnetic flux leakage when a field weakening current is applied, thus resulting in a larger d-axis inductance and lower maximum power at maximum speed when the motor voltage is limited by a battery or other means. This implies that the magnet arrangement is suitable for applications that require higher torque characteristics at low speeds.

Next, the case of a ▽-type rotor is described. The ▽-type

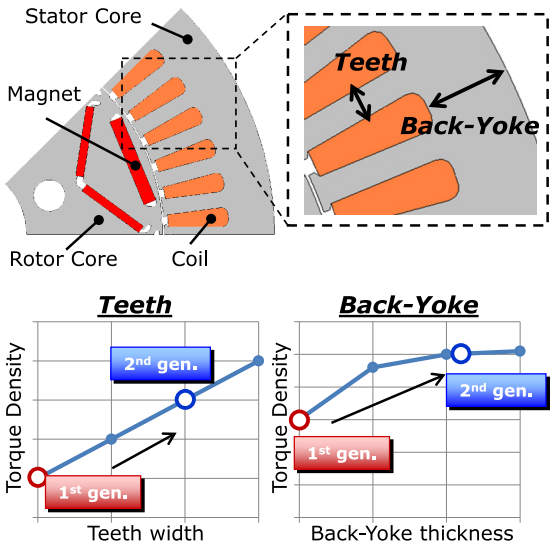


Fig. 17. Stator shape optimization

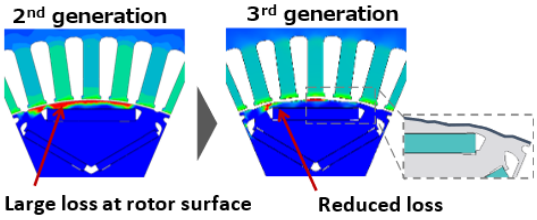


Fig. 18. Rotor shape optimization

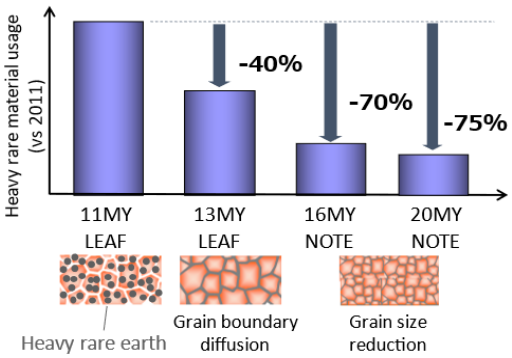


Fig. 19. HREE quantity progress

rotor has a two-layer magnet arrangement structure, which makes it possible to secure a q-axis magnetic path between the magnet layers and achieve a maximum torque comparable to that of a V-type rotor. Additionally, because the outer magnets are arranged on the outside of the rotor, the d-axis inductance can be kept small because the electromagnetic steel in this area can be reduced. As a result, higher power can be obtained even when the motor voltage is limited. Therefore, the ▽-type arrangement is suitable for EV motors that require both torque and power characteristics. Nissan installed a first-generation ▽-type rotor motor in the world's first mass-produced EV, the 11MY LEAF.

The second-generation motor for the 13MY LEAF retained the concept of the first generation. Specifically, we adopted the measures to reduce the excitation force described in the previous section and increased the width of the teeth and back yoke to secure the magnetic path area in the motor cross section, as shown in Figure 17, thereby reducing the thickness of the motor core by 10 mm without deteriorating the torque constant.

Furthermore, as mentioned earlier, the third-generation motor was designed specifically for compact cars.

Because the motor is designed for compact cars and does not require high maximum power, the number of turns was changed from 9 to 11, thus reducing the maximum power, and prioritizing the maximum torque, resulting in a shortened motor core length of 115 mm. Additionally, to improve the motor efficiency, the iron loss on the outermost surface of the rotor, which occurred in the second generation, was significantly reduced while maintaining high noise and vibration performance by creating a convex and concave surface on the rotor surface. (Figure 18)

Finally, we discuss the reduction of heavy rare earth elements (HREE) such as Dy, Tb in the magnets. HREE are elements that increase the heat resistance of neodymium magnets. However, because their production is limited, it is necessary to reduce their consumption as much as possible to ensure a stable supply for the future expansion of electric vehicles. Figure 19 shows the amount of HREE used as a ratio of 11MY LEAF to 100%. The development of magnet manufacturing technologies, such as grain boundary diffusion technology and fine crystallization, has significantly reduced the amount of HREE used in magnets with the same heat resistance performance. Additionally, to suppress the temperature rise of the magnet, the above-mentioned losses have been reduced, and in the 20MY NOTE, the amount of HREE has been reduced by 75% compared to the 11MY LEAF.

As described above, at the time of the first FY11 LEAF, we adopted the basic structure of a motor with good features. Moreover, through repeated improvements in the magnetic circuit design, we successfully reduced the size and cost of the motor, resulting in a highly competitive motor.

5. Evolution of Inverter

With each new generation the size of inverter becomes smaller. The most important components that determine the size, performance, and cost of an inverter are the power semiconductors and the power modules that package them for cooling.

The total chip area of insulated gate bipolar transistor (IGBT) and diodes has been dramatically reduced while output current has increased from the first to the third generation. As a result, the current density per chip size has increased more than four times (Table 5). In addition to reducing the losses by improving the performance of the power semiconductors themselves, the following technologies have made this possible.

(1) Reducing thermal resistance by improving the cooling structure

(2) Maximum utilization of power semiconductor performance through temperature control

(3) Reducing the size of printed circuit boards by optimizing circuit design and improving mounting density.

Particularly, technology (3) is necessary to leverage smaller power modules in the miniaturization of inverters. In addition to these technologies, the integration of other components and layout optimization have enabled the 20MY

Table 5. Inverter specifications

	EV				e-POWER			
	11MY LEAF	13MY LEAF	18MY LEAF	LEAF e+	SERENA	16MY NOTE	20MY NOTE	AURA
	1st generation		2nd generation				3rd generation	
INV Mass [kg]	16.7		15.3		16.3		10.1	
INV Vol [L]	11.8		13.2		13.9		8.4	
PM size [mm x mm]	227x193		213x97				98x100	
Max current [Arms]	400	360	500			360	500	
IGBT size [mm^2]	675	435	398			201	216	
FWD size [mm^2]	591	378	200			133	152	
Chip current density ratio	Base	Base	2.6			3.4	4.3	
PM cooling	Indirect cooling		Direct cooling				Direct cooling	

NOTE to reduce the volume of the inverter by −40% compared to the previous generation 16MY NOTE.

The following sections explain the characteristics of each of the technologies from (1) to (3), as shown below.

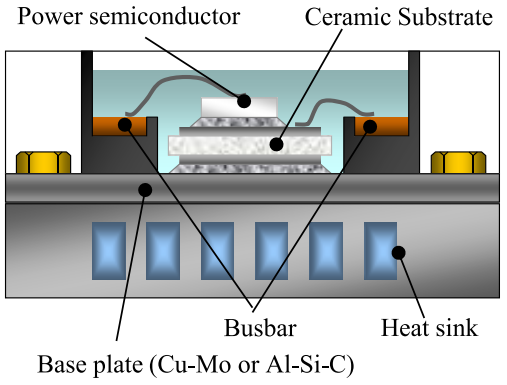
5.1 Reducing Thermal Resistance by Improving Cooling Structure When the first LEAF was developed, there were very few power modules for automotive applications, and it was difficult to find products that met the expected cost and reliability.

Therefore, Nissan manufactured a power module in-house with an original structure that combines both low cost and reliability.

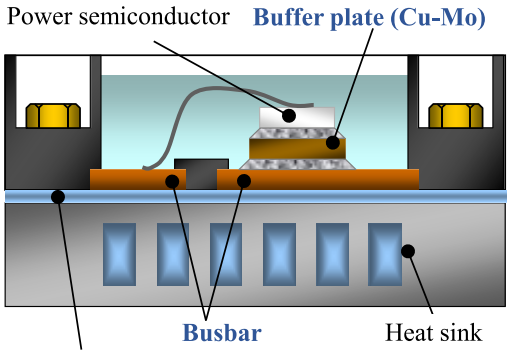
Generally, power modules have a structure in which power semiconductor chips are mounted on expensive insulating substrates such as ceramics and connected to busbars by wire bonding to draw current. This structure used expensive insulating substrates, had a small heat dissipation area, and poor heat dissipation performance. (Figure 20(a)) In the first-generation LEAF, the expensive insulating substrate was eliminated, and the power semiconductor chips were mounted directly on the busbars facing the water channel. Furthermore, to improve cooling performance, the busbars were cooled through a high thermal conductivity insulating film. (Figure 20(b)).

The challenge with this structure is that the difference in linear expansion coefficients between the semiconductor chip and the busbar results in greater stress on the solder. The solution is to place a buffer plate between the semiconductor chip and the busbar, which has an intermediate linear expansion coefficient between the two. While a typical power module requires a large ceramic substrate that occupies the entire bottom of the package, this buffer plate is only the size of a chip, thus reducing the cost.

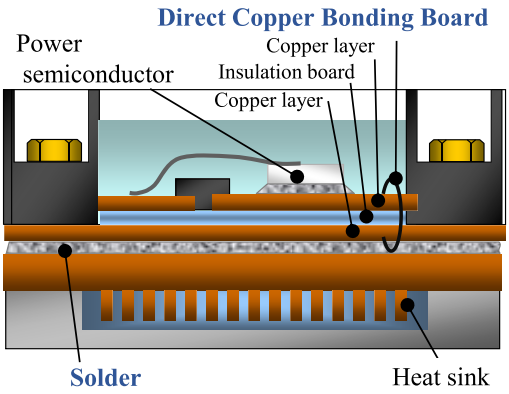
When beginning the development of the 18MY LEAF and the first-generation e-POWER, semiconductor suppliers had established a comprehensive power module supply system for automotive applications, and Nissan switched from in-house production to purchasing from semiconductor suppliers. At that time, the indirect cooling structure of the power modules was common, in which water is cooled by adhering the module to a heat sink through grease, etc. However, to reduce thermal resistance, this second-generation inverter adopted a direct cooling structure with high-efficiency fins integrated into the power module. (Figure 20(c)) The power module of the third-generation inverter used in the second-generation e-POWER has copper leads that are bonded directly to the surface of the semiconductor chip to draw current, which further reduces thermal resistance in addition to saving space. This reduction in thermal resistance, along with the improved performance of the semiconductors themselves, has enabled third-generation inverters to deliver the required current with only one IGBT chip per channel. As a result, the size of power modules has been dramatically reduced.



(a) Power module of conventional inverter



(b) Power module of 1st Gene. inverter



(c) Power module of 2nd Gene. inverter

Fig. 20. Power module vertical structure

5.2 Maximize the Performance of Power Semiconductors by Controlling Temperature To determine the size of a power semiconductor chip, in addition to the loss characteristics of the semiconductor itself and the thermal resistance of the package, it is important to control the temperature of the semiconductor and maintain a margin above the critical temperature.

Generally, the semiconductor temperature is controlled based on the information from temperature sensors embedded in the semiconductor chips or package. In this case, it

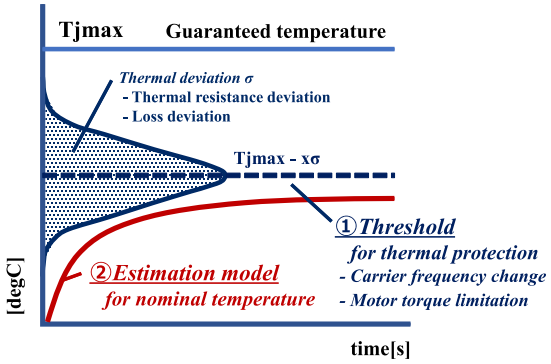


Fig. 21. Concept of power semiconductor thermal protection

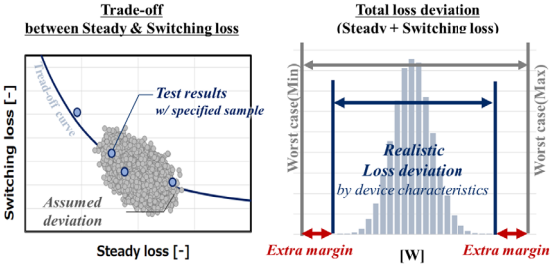


Fig. 22. Loss deviation of power semiconductor

is necessary to control the semiconductor temperature with the necessary margin from the allowable semiconductor temperature because the response delay of the sensor and error factors must be considered. Therefore, either a large margin must be applied to the semiconductor chip size or excessive operating restrictions must be imposed.

Nissan has been able to reduce the size of power semiconductor chips without requiring excessive protection even in complex vehicle behavior by controlling temperature based on the following two concepts. (Figure 21)[11]

a. Temperature protection threshold with margin based on accurately identified variation factors

Losses are caused by the electrical characteristics of semiconductors, and variation is quantified by understanding the correlation between the key characteristics and losses. The characteristics with trade-offs are quantified by considering the range of variation that can realistically occur, instead of a simple combination of the worst values of each. (Figure 22)

Thermal resistance, the other factor, is caused by the cooling structure of the power module. Therefore, the physical structure in the vertical direction from the semiconductor junction to the coolant is modeled, and the amount of variation is quantified by considering the manufacturing variation factors and the change in thermal resistance due to aging degradation. (Figure 23)

Finally, the temperature protection threshold is set by subtracting the rise in temperature, which considers the abovementioned accurately quantified loss and thermal resistance variations, from the rated temperature of the power semiconductor as a margin.

b. Temperature estimation method using thermal resistance model with variable parameters depending on operating conditions

The temperature estimation model consists of a thermal resistance model and a loss model.

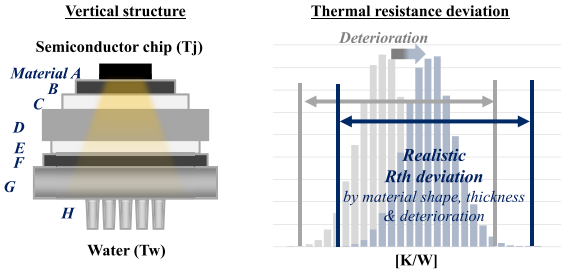


Fig. 23. Thermal resistance deviation of powe module

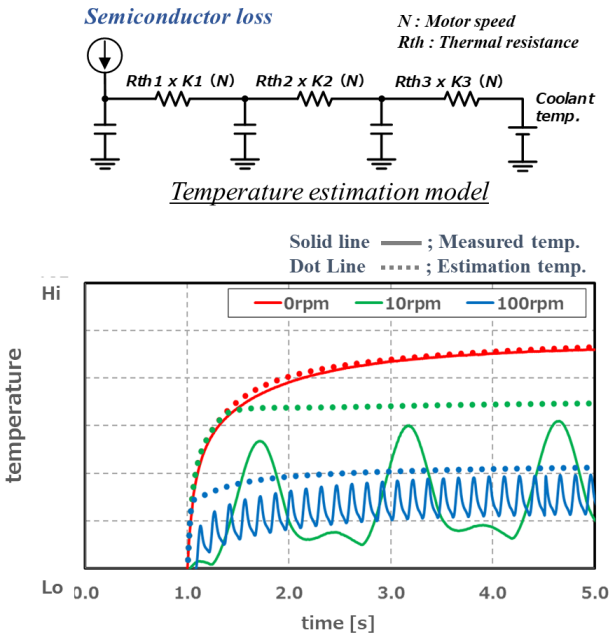


Fig. 24. Temperature estimation model and its result

The thermal resistance model is based on the abovementioned cooling structure model and is a multi-stage combination of first-order time delay. In an actual vehicle, the peak temperature transition of the chip varies according to the thermal time constant of the heat dissipation system because the energization interval to the power semiconductor chip for each phase varies due to various operating conditions such as motor lock, very low speed, and high-speed rotation. Nissan replaces this motion with changes in the apparent thermal resistance due to operating conditions and reproduces it by setting variable gain parameters for the aforementioned multiple time delays of first-order systems according to rotational speed. For the loss model, Nissan uses an approximate equation in which the control parameter are the inverter operating conditions, such as control current and battery voltage, and obtained loss characteristics, which consider the variation of the generated loss depending on the temperature of the semiconductor devices. (Figure 24)

5.3 PCB Downsizing by Optimizing Circuit Design and Improving Component Density To maximize the effect of downsizing the power module on the inverter, it is necessary to downsize the circuit board to be combined with the power module. Particularly, the following approaches are used in the design of second and third generation inverters.

- Improvement of the mounting density using small chip

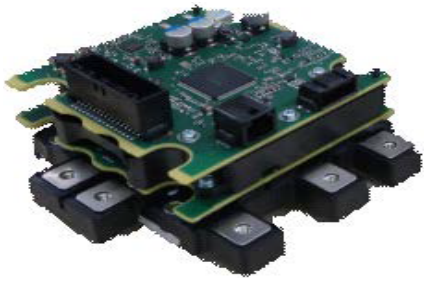


Fig. 25. Inverter module (3rd generation inverter)

components and build-up board.

- Reduction of the number of components by integrating functions using ICs and centralized power supplies.
- Reduction of wiring harnesses using board-to-board connectors

The circuit board has been downsized to the same projected area as the power module, and the "inverter module" integrated with the power module is compact enough to fit in the palm of the hand. (Figure 25)

6. Conclusion

Nissan developed the LEAF as a mass-produced EV in 2010 and the NOTE with the 100% electric drive system e-POWER in 2016. Since then, Nissan has made continuous improvements to deliver the value of 100% electric drive to customers.

The motor and inverter, which are the main components of this system, have been made smaller and lighter by integrating the mechanical and electrical components, while solving problems such as heat and vibration to improve vehicle install ability.

Motors have been reduced in size by reducing losses by tuning magnetic circuits, etc., while maintaining the basic specifications such as outer diameter, and the use of heavy rare earth elements has been significantly reduced by the advances in magnetic materials.

In the inverter, Nissan has improved the cooling performance in the vertical structure of the power module, and has applied the unique temperature prediction method to fully utilize semiconductors potential even under the complex operating conditions of automobiles. Together with the reduction of losses in the semiconductors themselves, Nissan has dramatically downsized the inverter.

Nissan will continue its efforts to further reduce size and weight to bring the value of Nissan's 100% electric drive to more customers and contribute to the early realization of carbon neutrality.

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